

Example of Fastforward attack without padding

$$p = 5 \quad q = 11$$

$$n = 55 - \text{public}$$

$$e = 3 - \text{public}$$

$$d = 27 - \text{private}$$

- Forge signature without knowing d
- They simply pick a random s e.g. $s = 2$
- Compute:

$$m = s^e \bmod n$$

$$m = 2^3 \bmod 55$$

$$m = 8$$

Present $(m = 8, s = 2)$ as forged message/signature pair

Verification:

$$s^e \bmod n = 2^3 \bmod 55 \\ = 8$$

matches m